
Do Reward Hacking Guardrails Reduce Exploration Diversity in GRPO?

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Abstract

Reinforcement learning with verifiable rewards, particularly Group Relative Policy Optimization (GRPO), has become a standard approach for training reasoning models. Proxy reward signals, such as length bonuses, create incentives for reward hacking, and practitioners commonly respond with guardrails such as KL regularization or hard length caps. Whether these guardrails change the diversity of a policy’s reasoning strategies, independent of their effect on task correctness, has not been systematically measured. We study this question using Qwen2.5-1.5B-Instruct fine-tuned with LoRA on the Countdown arithmetic task, across seven conditions: a clean baseline, a hackable length-bonus reward, three KL-regularized variants at $\beta \in \{0.01, 0.05, 0.10\}$, a hard length cap, and a combined KL-plus-cap guardrail. The six core conditions and the combined condition are each trained with three random seeds (42, 123, 456); a length-cap severity sweep at four cap values (45, 65, 100, 128 tokens) uses a single seed and is reported as preliminary. We measure diversity through pass@k, the pass@k minus pass@1 exploration gap, unique solution count, and the embedding variance of sampled reasoning traces. Our most robust result is that KL regularization increases semantic diversity monotonically with β : embedding variance moves from 0.134 (baseline) to 0.134 ($\beta = 0.01$) to 0.246 ($\beta = 0.05$) to 0.286 ($\beta = 0.10$), with tight seed agreement at the two higher coefficients and a corresponding cost to greedy accuracy. An exact Jonckheere-Terpstra trend test across all 12 observations (3 seeds $\times$ 4 KL levels) confirms this monotone ordering (one-sided $p = 0.0009$), as does the corresponding pass@1 decline ($p = 0.0005$). Hard length caps show a qualitatively different and less reliable pattern: two of three seeds internalize the cap and collapse embedding variance to roughly a quarter of the baseline value, but the third seed fails to internalize the constraint, producing high length variance and elevated embedding variance instead, a bimodal outcome rather than a single consistent effect. Combining a moderate KL penalty with a hard cap reproduces KL-only levels of diversity (embedding variance 0.282) at roughly half the median reasoning length, consistently across all three seeds, though at a correctness cost comparable to the cap-alone condition. Embedding variance and unique solution count, our two diversity metrics, are themselves largely uncorrelated across the seven conditions and diverge sharply within the KL series, where embedding variance rises as unique solution count falls; at matched embedding variance, the combined guardrail nonetheless recovers a markedly higher unique solution count and pass@8 than the high-KL condition alone, without leading the full table on unique solution count. Reward hacking through length exploitation alone leaves diversity within seed noise of the clean baseline while still lowering greedy accuracy. Together these results show that the specific mechanism of a guardrail, not the presence of reward hacking itself, determines whether a trained policy stays exploratory or collapses onto a narrow strategy, and that this outcome can itself depend on the training seed rather than being guaranteed by the guardrail alone.

1 Introduction

The emergence of reasoning models trained with reinforcement learning from verifiable rewards has marked a significant advance in large language model capabilities. DeepSeek-R1 [DeepSeek-AI, 2025] demonstrated that Group Relative Policy Optimization (GRPO) [Shao et al., 2024] can effectively train models to produce structured reasoning traces, achieving strong performance on mathematical and logical reasoning tasks. The core idea is elegant: rather than relying on human preference judgments, the model learns from environments where correctness can be automatically verified, such as arithmetic computation or code execution.

However, this paradigm introduces a well-known vulnerability: reward hacking [Skalse et al., 2022, Gao et al., 2023]. When the reward signal includes proxy components, such as length bonuses meant to encourage verbose reasoning, the policy can exploit these signals without improving genuine reasoning quality. A model trained with a length bonus may learn to produce increasingly long, meandering reasoning traces that receive high reward despite being no more (or even less) likely to reach correct answers.

Practitioners have responded with guardrails designed to constrain this hacking behavior. Two common approaches are *KL regularization*, which penalizes divergence from a reference policy [Ziegler et al., 2019, Ouyang et al., 2022], and *hard environmental constraints*, such as maximum token limits that zero out the reward for overly long generations. These guardrails are typically evaluated on their ability to suppress reward hacking and maintain correctness.

But a question remains under-examined: **what effect do these guardrails have on exploration diversity?** In reinforcement learning, the exploration-exploitation tradeoff is fundamental [Sutton and Barto, 2018]. A policy that converges too quickly to a narrow reasoning strategy may miss better solutions; a policy that explores too broadly may fail to execute reliably. Guardrails, by constraining the policy’s behavior, implicitly shift this balance. Prior work has not systematically measured how different guardrail designs affect the diversity of reasoning strategies learned by GRPO-trained models, and, as we show, a single training run per condition is not enough to tell a reliable effect from a seed-dependent accident.

We address this gap through a controlled experiment on the Countdown arithmetic reasoning task. We train Qwen2.5-1.5B-Instruct with LoRA across seven conditions that isolate the effects of reward hacking, KL regularization, hard length constraints, and their combination, with three seeds per condition for the six core conditions and the combined condition. Our key findings are:

- **KL regularization monotonically increases semantic diversity, and this is our most consistent result.** Embedding variance rises from 0.134 at baseline to 0.286 at $\beta = 0.10$, with the two higher β values showing tight agreement across all three seeds, at a monotonic cost to greedy accuracy (exact JT trend test, $p = 0.0009$ across 12 observations).
- **Hard length caps produce a bimodal, seed-dependent outcome rather than a single reliable effect.** Two of three seeds collapse embedding variance to about a quarter of baseline once the cap is internalized; a third seed fails to internalize the cap and instead shows high length variance and diversity closer to an unconstrained run.
- **Combining a moderate KL penalty with a hard cap reliably reproduces KL-level diversity at much shorter reasoning length.** Across all three seeds, the combined condition reaches embedding variance of 0.282, comparable to the high-KL condition alone (0.286), at roughly half the median token length, though without a correctness advantage over the cap-alone condition.
- **Reward hacking via length exploitation does not clearly reduce semantic diversity.** The hackable condition’s embedding variance (0.155) is within seed noise of the clean baseline (0.134) despite roughly tripling median reasoning length, though it does cost greedy accuracy.

Our contributions are fivefold:

1. We present a multi-seed empirical study of guardrail effects on exploration diversity in GRPO-trained reasoning models, extending a single-seed pilot to three seeds for six conditions plus a combined-guardrail condition.
2. We show that hard length caps do not reliably collapse diversity: the effect depends on whether the training seed leads the policy to internalize the constraint, a distinction invisible in an aggregate mean.

3. We identify KL regularization strength as the most seed-robust lever on exploration diversity in our setting, and show that combining it with a hard cap can reproduce most of its diversity benefit at substantially lower reasoning length.
4. We reframe the practitioner’s choice of guardrail as a decision with measurable, and sometimes unreliable, consequences for exploration, rather than a single knob that trades off against correctness alone.
5. We show that embedding variance and unique solution count are largely uncorrelated across our seven conditions, diverge sharply and near-monotonically as a function of KL strength, and that the combined guardrail recovers unique-solution-count and pass@8 levels closer to baseline while matching high-KL embedding variance, without leading the full table on unique solution count.

2 Background and Related Work

2.1 Group Relative Policy Optimization

GRPO [Shao et al., 2024] is a reinforcement learning algorithm designed for training language models with verifiable rewards. Unlike PPO [Schulman et al., 2017], GRPO does not use a critic network; instead, it normalizes rewards within a group of generations for the same prompt. For each prompt x , the model generates G completions $\{y_1, \dots, y_G\}$, and the advantage for completion y_i is computed as $A_i = (r_i - \mu_r) / \sigma_r$, where μ_r and σ_r are the mean and standard deviation of rewards within the group. The policy is updated to increase the probability of high-advantage completions while a KL penalty constrains deviation from the reference policy. DeepSeek-R1 [DeepSeek-AI, 2025] demonstrated the effectiveness of this approach for training reasoning models at scale.

2.2 Reward Hacking in Language Models

Reward hacking occurs when a policy exploits proxy reward signals to achieve high reward without fulfilling the intended objective [Skalse et al., 2022]. In language model training, common proxy signals include response length, use of formatting tokens, or specific lexical patterns. Gao et al. [2023] showed that reward models trained on human preferences can be exploited by generating overly long or stylistically distinctive responses. In the GRPO setting, length bonuses are particularly problematic because they create a direct incentive for the model to produce longer reasoning traces, regardless of whether additional length improves reasoning quality.

2.3 KL Regularization in RLHF

KL regularization was introduced in the context of RLHF to prevent the policy from diverging too far from the supervised fine-tuned model [Ziegler et al., 2019, Ouyang et al., 2022]. The standard objective adds a term $-\beta \cdot D_{\text{KL}}(\pi_\theta \parallel \pi_{\text{ref}})$ to the reward, where β controls the strength of regularization. In InstructGPT [Ouyang et al., 2022], KL regularization was shown to improve training stability and prevent reward hacking. Its effect on exploration diversity, and how reliably that effect holds across training seeds, has not been systematically studied.

2.4 Exploration in Reinforcement Learning

The exploration-exploitation tradeoff is a foundational concept in reinforcement learning [Sutton and Barto, 2018]. In the context of LLM training, exploration corresponds to the diversity of generated responses, which affects both the discovery of better solutions and the robustness of the learned policy. Recent work has begun to examine exploration in LLM training. Chen et al. [2021] introduced the pass@ k metric for measuring the probability of generating at least one correct solution in k samples, which captures both correctness and diversity.

2.5 Related Empirical Studies

Several recent works touch on related themes. Ichihara et al. [2025] propose MO-GRPO, which normalizes reward components by their variances to prevent GRPO from over-optimizing a single objective at the expense of others in multi-objective settings; our work is complementary in that we focus on exploration diversity within a single scalar reward rather than on balancing multiple

Table 1: Experimental conditions. All conditions include correctness and format rewards. The hackable conditions add a length bonus. KL regularization and the hard length cap are applied as guardrails, alone or combined. C1–C6 and C7 use three seeds each; the cap severity sweep (Section 4.8) uses one seed and is reported separately.

Condition	Reward Components	KL β	Length Cap	Seeds
C1: Baseline	Correctness + Format	0	None	3
C2: Hackable	Correctness + Format + Length Bonus	0	None	3
C3: KL-Low	Correctness + Format + Length Bonus	0.01	None	3
C4: KL-Med	Correctness + Format + Length Bonus	0.05	None	3
C5: KL-High	Correctness + Format + Length Bonus	0.10	None	3
C6: LengthCap	Correctness + Format + Length Bonus	0	65 tokens	3
C7: Combined	Correctness + Format + Length Bonus	0.05	65 tokens	3

objectives. Çağatan and Zhao [2026] revisit AI safety gridworlds and find that language model agents from 1.5B to 14B parameters exhibit zero-shot specification gaming, where observed reward and hidden safety objectives diverge, and that standard mitigations including exploration prompts and entropy regularization fail to close this gap; their setting is interactive gridworlds rather than the static reasoning task we study, but their negative result for entropy-based mitigation is a useful point of contrast for our KL-based findings. Le et al. [2026] propose Transformation-Augmented GRPO, which pools responses across paraphrased problem variants specifically to counter what they call diversity collapse in GRPO training; we do not modify the training procedure to enhance exploration, but instead measure how existing guardrail choices affect the same phenomenon they target.

3 Experimental Setup

3.1 Task and Dataset

We use the Countdown arithmetic reasoning task, where the model is given a set of numbers and a target value, and must produce an arithmetic expression using $+$, $-$, $\times$, and $\div$ that evaluates to the target, using each number exactly once. We use the Jiayi-Pan/Countdown-Tasks-3to4 dataset from HuggingFace [Pan, 2024], which contains problems with 3–4 numbers. We hold out 50 evaluation examples with a fixed random seed, independent of the training seed, for reproducible evaluation across all conditions.

3.2 Model and Training

We fine-tune Qwen2.5-1.5B-Instruct using LoRA with rank $r = 16$, $\alpha = 32$, and dropout 0.05 applied to all projection layers. Training runs for 1000 steps with batch size 1, gradient accumulation 4, learning rate 1×10^{-5} , and cosine learning rate schedule. During GRPO training, we generate 8 completions per prompt for advantage computation.

3.3 Experimental Conditions

We design seven conditions to isolate the effects of reward hacking, KL regularization, hard length constraints, and their combination. Table 1 summarizes the conditions and the number of training seeds used for each. The six core conditions (C1–C6) and the combined guardrail condition (C7) are each trained with three seeds (42, 123, 456). A separate length-cap severity sweep, described in Section 4.8, trains four additional cap values at a single seed (456) and is treated throughout as a preliminary, single-seed result rather than a seed-validated finding.

3.4 Evaluation Metrics

We define five metrics to capture different aspects of exploration diversity:

pass@1: The proportion of evaluation examples where greedy decoding (temperature 0) produces a correct expression. This measures the policy’s ability to reliably execute its best strategy.

pass@k ($k = 8$): The probability that at least one of k sampled completions is correct, estimated using the unbiased estimator from Chen et al. [2021]:

$$\text{pass@k} = \mathbb{E}_{\text{problems}} \left[1 - \frac{\binom{n-c}{k}}{\binom{n}{k}} \right]$$

where $n = 8$ samples are drawn per problem and c is the number of correct samples.

Exploration gap: The difference $\text{pass@k} - \text{pass@1}$. A larger gap indicates that the policy explores diverse strategies, some of which succeed even when the greedy strategy fails.

Unique solution count: The proportion of distinct correct expressions among k samples, averaged across problems. This measures syntactic diversity of correct solutions. As in the results below, this metric shows high seed variance for several conditions and should be read alongside embedding variance rather than in isolation.

Embedding variance: The mean pairwise cosine distance between sentence-transformer embeddings of the <think> block contents. This measures semantic diversity of the reasoning traces. Higher values indicate more diverse reasoning strategies.

3.5 Seeds, Reproducibility, and Statistical Reporting

The six core conditions and the combined guardrail condition (C7) are each trained with three random seeds (42, 123, 456). Evaluation uses a fixed seed, independent of the training seed, for all sampling-based metrics to ensure reproducibility. All main-text results are reported as mean $\pm$ sample standard deviation ($n - 1$ denominator) across the three seeds unless otherwise noted. The length-cap severity sweep (Section 4.8) uses a single seed (456) and is reported without a standard deviation, as a preliminary result.

With only three seeds per condition, all significance tests reported in this paper are exact and non-parametric; we do not use parametric tests anywhere. At $n = 3$ per group, exact permutation tests can achieve a minimum two-sided p -value of 0.10 ($\binom{6}{3} = 20$ relabelings); no result can reach conventional significance ($p < 0.05$) under this design regardless of effect size. We therefore report effect sizes (Cohen’s d) alongside p -values throughout, and treat effect size as the primary interpretive quantity. BCa bootstrap confidence intervals over the 3 seeds are reported in Table 3 as rough uncertainty sketches; they should not be read as precise inferential bounds at this sample size. The length-cap severity sweep (Section 4.8) uses a single seed (456) and is reported without inferential statistics, as a preliminary result. Full discussion of these statistical limitations appears in Appendix B.

4 Results

4.1 Main Results

Table 2 presents the three-seed aggregate results across all seven conditions. Per-seed raw data is shown in Appendix A.

Table 2: Main evaluation results across all conditions. Best value in each column is bolded. All metrics are mean $\pm$ sample standard deviation across 3 seeds (42, 123, 456), except Med. Len., which is the mean of the three per-seed medians. [†]C6’s embedding variance mean is not bolded or read as a single effect: two of three seeds collapse to ~ 0.03 while one seed reaches 0.171, see Section 4.3.

Condition	pass@1	pass@8	Gap	Unique	Emb. Var.	Med. Len.
C1: Baseline	0.21 $\pm$ 0.04	0.40 $\pm$ 0.08	0.19 $\pm$ 0.08	0.49 $\pm$ 0.10	0.134 $\pm$ 0.017	38.7
C2: Hackable	0.15 $\pm$ 0.01	0.43 $\pm$ 0.04	0.27 $\pm$ 0.03	0.58 $\pm$ 0.11	0.155 $\pm$ 0.068	134.5
C3: KL-Low	0.13 $\pm$ 0.02	0.39 $\pm$ 0.06	0.26 $\pm$ 0.09	0.51 $\pm$ 0.06	0.134 $\pm$ 0.061	139.7
C4: KL-Med	0.13 $\pm$ 0.03	0.32 $\pm$ 0.03	0.19 $\pm$ 0.02	0.40 $\pm$ 0.03	0.246 $\pm$ 0.023	107.8
C5: KL-High	0.07 $\pm$ 0.03	0.27 $\pm$ 0.04	0.20 $\pm$ 0.02	0.31 $\pm$ 0.08	0.286 $\pm$ 0.007	78.3
C6: LengthCap	0.15 $\pm$ 0.07	0.33 $\pm$ 0.05	0.17 $\pm$ 0.04	0.27 $\pm$ 0.19	0.079 $\pm$ 0.080 [†]	59.0
C7: Combined	0.15 $\pm$ 0.03	0.39 $\pm$ 0.06	0.25 $\pm$ 0.09	0.45 $\pm$ 0.14	0.282 $\pm$ 0.010	40.3

Table 3: BCa bootstrap 95% confidence intervals for embedding variance and pass@1, bootstrapped over 3 seeds with 10,000 resamples. At $n = 3$ these intervals are rough uncertainty sketches; CI overlap or separation should be corroborated by the permutation tests in Section 4.4 before being read as a significance indicator. See Appendix B for full statistical limitations.

Condition	Mean Emb. Var.	95% BCa CI	Mean pass@1	95% BCa CI
C1: Baseline	0.134	[0.123, 0.144]	0.213	[0.180, 0.240]
C2: Hackable	0.155	[0.077, 0.197]	0.153	[0.140, 0.160]
C3: KL-Low	0.134	[0.079, 0.174]	0.127	[0.100, 0.140]
C4: KL-Med	0.246	[0.221, 0.260]	0.127	[0.100, 0.147]
C5: KL-High	0.286	[0.280, 0.290]	0.073	[0.040, 0.093]
C6: LengthCap	0.079	[0.026, 0.127]	0.153	[0.080, 0.220]
C7: Combined	0.282	[0.273, 0.288]	0.147	[0.120, 0.167]

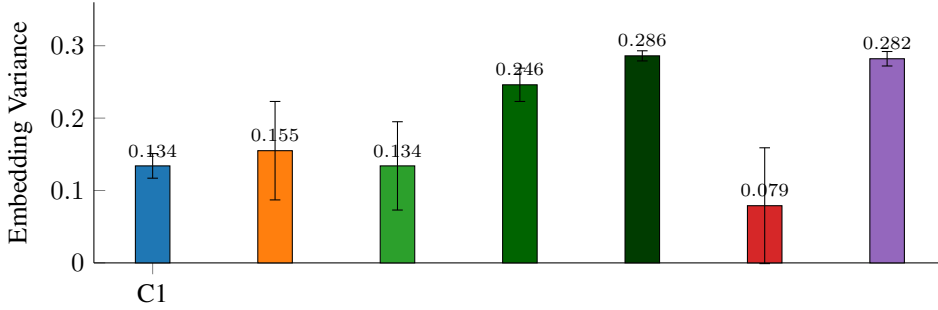


Figure 1: Embedding variance across all seven conditions, mean $\pm$ sample standard deviation over 3 seeds. KL-regularized conditions (C3–C5, green shades) show increasing diversity with β . C6’s wide error bar reflects a bimodal outcome across seeds, not uniform moderate diversity, see Section 4.3. C7 (purple) matches C5’s diversity at less than half its median reasoning length.

We verified every mean and standard deviation in Table 2 directly from the nine (three-seed $\times$ six core conditions) and three (C7) per-seed rows in Appendix A; no arithmetic discrepancies were found against the source logs.

4.2 Finding 1: Reward Hacking Does Not Clearly Reduce Semantic Diversity

Comparing C2 (hackable, no guardrail) to C1 (clean baseline), reward hacking successfully exploits the length bonus: median reasoning length roughly triples, from 38.7 to 134.5 tokens. Embedding variance, by contrast, moves only from 0.134 ± 0.017 to 0.155 ± 0.068 . Given a seed spread that ranges from 0.077 to 0.205 within C2 alone, this difference is well within seed noise; we do not read it as evidence that hacking preserves diversity exactly, only that the point estimate is comparable to baseline once seed variance is accounted for.

Finding 1. *Reward hacking via length exploitation does not clearly reduce semantic diversity of reasoning traces. Despite roughly tripling median reasoning length, the hackable policy’s embedding variance is comparable to the clean baseline within seed noise.*

One plausible explanation is mechanistic: length expansion increases the *budget* available for reasoning without constraining its *content*. The model learns to write more tokens, but nothing in the length bonus rewards converging on a single template. This is consistent with, but not proven by, our data. The correctness cost is more clear cut and holds across all three seeds: pass@1 falls from 0.21 ± 0.04 (baseline) to 0.15 ± 0.01 (hackable), with C2’s own seed variance on pass@1 markedly lower than its variance on embedding variance.

4.3 Finding 2: Hard Length Caps Produce a Bimodal, Seed-Dependent Outcome

This is the most substantially revised finding relative to the single-seed pilot version of this study, which reported a uniform 79% diversity collapse under the hard cap as its most robust result. With three seeds, C6’s embedding variance is 0.079 ± 0.080 , a standard deviation roughly equal to the

mean. This is not noise scattered around one number; it is two distinct outcomes, shown per seed in Table 4.

Table 4: C6 (hard length cap, 65 tokens) per-seed breakdown. Seeds 42 and 123 internalize the cap; seed 456 does not. Len. Std is the standard deviation of sampled completion length (reasoning_length/std from the full evaluation log). *Seed 42’s value is an approximate figure carried from an earlier summary, not a logged full-precision statistic; seeds 123 and 456 are exact.

Seed	pass@1	pass@8	Unique	Emb. Var.	Med. Len.	Len. Std
42	0.22	0.38	0.06	0.026	64.0	~6*
123	0.16	0.30	0.40	0.040	61.0	6.63
456	0.08	0.30	0.36	0.171	52.0	26.32

Seeds 42 and 123 show embedding variance of 0.026 and 0.040, with reasoning length standard deviation (reasoning_length/std from the full evaluation log, computed across sampled completions within each run) of approximately 6 tokens for seed 42 and exactly 6.63 tokens for seed 123 (mean length 61.34, median 61.0, range 44 to 88); the cap is internalized in both, and diversity collapses to roughly a quarter of the baseline value. Seed 456 shows embedding variance of 0.171 and a reasoning length standard deviation of 26.3 tokens (against the same 65-token cap, with a maximum observed length of 216 tokens in the length-cap severity sweep run at the same cap value, Section 4.8). The reward function still assigns zero reward to any completion whose <think> block exceeds the cap, so the training objective itself did not change for this seed. One plausible explanation, consistent with but not established by our data, is that at evaluation the seed-456 policy frequently fails to reproduce the compressed pattern reliably enough to stay under the cap, so its sampled completions span both compliant short answers and non-compliant long ones, inflating both length variance and, in turn, embedding variance.

Finding 2. *Hard length caps do not have a single, reliable effect on semantic diversity. When the cap is internalized during training, diversity collapses substantially; but internalization is not guaranteed, and failed in one of three seeds in this study. Seed-dependence is itself part of the finding, not noise to be averaged away.*

Reporting only the three-seed mean would hide this. We show both the aggregate (Table 2) and the per-seed breakdown (Table 4) because the aggregate understates how consistently the cap works when it works, and overstates how consistently it works overall.

A formal test of the C6 bimodality is not meaningful at this seed count. Treating seeds 42 and 123 as one group (mean embedding variance 0.033) against seed 456 alone (0.171) yields only $\binom{3}{1} = 3$ possible relabelings, giving a minimum achievable two-sided p of 0.333, the test has no power to distinguish a true bimodal effect from chance. We report this split descriptively only. The bimodality claim rests on the per-seed breakdown in Table 4 and the mechanistic interpretation in Section 5.3, not on a hypothesis test.

4.4 Finding 3: KL Regularization Monotonically Increases Semantic Diversity, and Is Our Most Robust Result

Across three seeds, embedding variance rises with β : 0.134 (baseline) $\rightarrow$ 0.134 ($\beta = 0.01$) $\rightarrow$ 0.246 ($\beta = 0.05$) $\rightarrow$ 0.286 ($\beta = 0.10$). An exact Jonckheere-Terpstra trend test across all 12 observations confirms this ordering (one-sided $p = 0.0009$ for embedding variance; $p = 0.0005$ for pass@1 in the decreasing direction). Pairwise exact permutation tests on embedding variance between C1 and C5 yield $p = 0.100$, the minimum achievable in a 3-vs-3 design, where $\binom{6}{3} = 20$ total relabelings exist, with Cohen’s $d = -12.09$, indicating a very large effect. Between C5 and C7 (the diversity equivalence claim in Finding 7), $p = 0.600$ and Cohen’s $d = 0.53$, consistent with the near-zero mean difference of 0.004 and confirming that the combined guardrail reproduces KL-level diversity. Between C1 and C2 (Finding 1), $p = 0.600$ and Cohen’s $d = -0.42$, consistent with the seed-noise interpretation. At $n = 3$ per group the minimum achievable two-sided p is 0.10; effect size carries more interpretive weight than the p -value alone at this sample size, and all p -values should be read accordingly. Full bootstrap confidence intervals for all conditions are reported in Table 3.

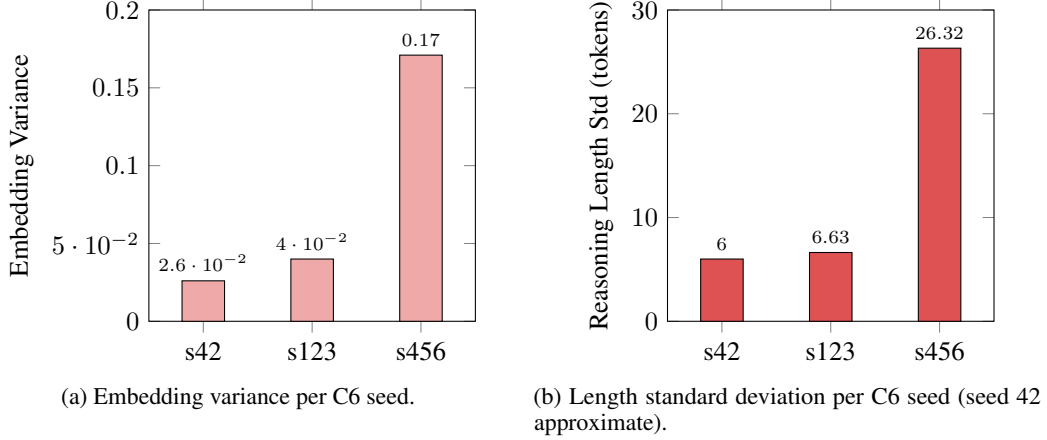


Figure 2: C6 per-seed bimodality. Seeds 42 and 123 internalize the 65-token cap (low embedding variance, low length variance); seed 456 does not (elevated on both). *Seed 42’s length-std value is an approximate figure carried from an earlier summary; seed 123 (6.63) is taken directly from its full evaluation log (reasoning_length/std), and seed 456 (26.32) from the matching cap-65 run in the severity sweep (Section 4.8).

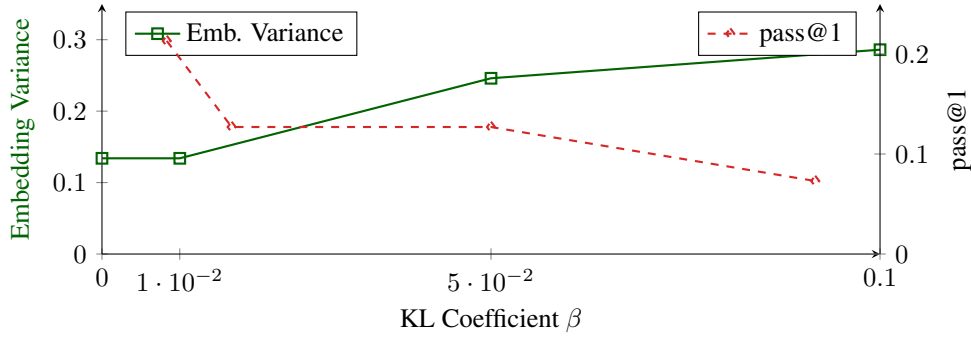


Figure 3: Exploration-exploitation tradeoff under KL regularization. Embedding variance (green, left axis) increases with β while pass@1 (red, right axis) decreases. Each point is the mean across 3 seeds.

C4 and C5 both show tight seed agreement, with standard deviation at 9% and 2% of their respective means; this is the most seed-robust pattern in the study once C7’s near-identical tightness is set alongside it (Section 4.7). C3 ($\beta = 0.01$) remains statistically indistinguishable from baseline in its mean (0.134 vs. 0.134) but carries one of the highest seed variances in the study (std 0.061, close to half its own mean), consistent with a below-threshold interpretation: at this coefficient, whether diversity increases at all appears to depend on the seed.

Finding 3. Across three seeds, KL regularization strength shows the most seed-robust relationship with semantic diversity we observe: embedding variance increases with β from 0.05 upward, with tight agreement across seeds at both $\beta = 0.05$ and $\beta = 0.10$. We treat this as the paper’s most confidently supported result.

The cost is equally clear, and close to monotonic: pass@1 falls from 0.21 ± 0.04 (baseline) to 0.13 ± 0.02 ($\beta = 0.01$) to 0.13 ± 0.03 ($\beta = 0.05$, statistically tied with $\beta = 0.01$) to 0.07 ± 0.03 ($\beta = 0.10$). KL regularization does not let the policy commit fully to any single reasoning strategy; the higher the coefficient, the more this holds, and the more consistently it holds across seeds.

This tradeoff is visualized in Figure 3. The dual-axis plot shows embedding variance increasing and pass@1 decreasing as β grows, using the three-seed means from Table 2.

4.5 Diversity Metrics Diverge Under KL Regularization, and the Combined Guardrail Recovers the Gap

The metrics defined in Section 3.4 are not interchangeable measures of the same underlying quantity. Table 5 reports Pearson correlations across the seven condition means (Table 2) for every pair among embedding variance, unique solution count, pass@8, and exploration gap.

Table 5: Pearson correlations across the seven condition means in Table 2. With $n = 7$, only two pairs reach $p < 0.05$.

Metric pair	Pearson r	p
Emb. Var. vs. Unique	-0.122	0.794
Emb. Var. vs. pass@8	-0.402	0.371
pass@8 vs. Unique	0.876	0.010
Gap vs. Unique	0.778	0.039
Gap vs. Emb. Var.	0.120	0.798

Embedding variance and unique solution count are near-orthogonal across the seven conditions ($r = -0.122$, not distinguishable from zero at this n), while unique solution count tracks pass@8 closely ($r = 0.876$, the only correlation in this set that reaches $p < 0.05$ alongside the gap-unique pair). This is a mechanistic consequence of how unique solution count is defined, not a coincidence: it only counts distinct correct expressions among the k sampled completions, so it is mathematically bounded above by how many of those completions are correct in the first place, which is exactly what pass@8 measures. Embedding variance is computed over the reasoning content of all sampled completions regardless of correctness, so it has no such bound. This is the reason embedding variance is treated as the paper’s primary diversity metric and unique solution count as a secondary, correctness-confounded one throughout this paper, a position stated informally in Section 3.4 and the limitations discussion; the correlation analysis here is the evidence for that position, not a new one.

The exploration gap correlates with unique solution count almost as strongly as unique correlates with pass@8 ($r = 0.778$, $p = 0.039$), which is evidence that gap is not a clean diversity measure either: since gap is pass@8 minus pass@1, and both terms move together under most guardrails in this study, gap ends up tracking overall correctness level almost as strongly as it tracks anything specifically exploratory. A policy that performs uniformly poorly at both greedy and sampled decoding can post a similar gap to one that is genuinely exploring, because both of its raw numbers are simply low.

Restricting to the four KL coefficients (C1 at $\beta = 0$, C3 at $\beta = 0.01$, C4 at $\beta = 0.05$, C5 at $\beta = 0.10$) makes this divergence sharpest. Table 6 shows embedding variance and unique solution count moving in nearly opposite directions as β increases.

Table 6: KL subgroup means (C1, C3, C4, C5). Embedding variance rises while unique solution count falls as β increases.

β	Condition	Emb. Var.	Unique
0.00	C1	0.134	0.49
0.01	C3	0.134	0.51
0.05	C4	0.246	0.40
0.10	C5	0.286	0.31

Within this subgroup, β vs. embedding variance gives Pearson $r = 0.963$ ($p = 0.037$), β vs. unique gives $r = -0.983$ ($p = 0.017$), and embedding variance vs. unique directly gives $r = -0.976$ ($p = 0.024$). Given $n = 3$ seeds per point, we corroborate this with pairwise seed-level concordance, comparing every seed in each β level against every seed in every higher β level (54 total pairwise comparisons across the four ordered groups): embedding variance increases in 48 of 54 pairs (88.9% concordant), and unique solution count decreases in 45 of 54 pairs (83.3% concordant). We present these percentages as informal corroboration in the same spirit as the Jonckheere-Terpstra trend test already reported for Finding 3 in Section 4.4, not as a new formal test with its own p -value, to avoid multiplying hypothesis tests without correction.

Finding 4. *Within the KL sweep, embedding variance and unique solution count move in nearly opposite directions as β increases: embedding variance rises monotonically (0.134 to 0.286, matching*

the trend already reported in Finding 3) while unique solution count falls in a near-mirrored pattern (0.487 to 0.313), with 88.9% and 83.3% pairwise seed concordance respectively. KL regularization does not simply increase diversity; it exchanges outcome diversity, how many different correct answers the policy finds, for process diversity, how varied its reasoning is regardless of outcome. The two are not the same quantity, and a guardrail that increases one can simultaneously decrease the other.

This raises the question of whether a guardrail can increase process diversity without paying for it in outcome diversity. C7 (the combined guardrail) offers a test case, evaluated here specifically against C5 rather than against the full table.

At an embedding variance statistically indistinguishable from the high-KL condition (C7: 0.282 ± 0.010 vs. C5: 0.286 ± 0.007 , Mann-Whitney $p = 0.507$, Cohen’s $d = -0.53$), the combined guardrail achieves a markedly higher unique solution count (0.453 vs. 0.313, Cohen’s $d = 1.24$, exact Mann-Whitney $p = 0.200$) and pass@8 (0.393 vs. 0.273, Cohen’s $d = 2.30$, exact Mann-Whitney $p = 0.100$). We report these effect sizes as the primary evidence, consistent with the paper’s existing statistical framing (Section 3.5, Appendix B) that treats effect size as more informative than p -value at $n = 3$. This is a comparison anchored specifically to C5, not a claim about the full table: C7’s unique solution count nonetheless ranks fourth of seven overall (0.45), behind C2 (0.58), C3 (0.51), and C1 (0.49). The claim is that C7 escapes the specific tradeoff visible within the pure KL series, matching C5’s process diversity while recovering some of the outcome diversity that the KL series gives up, not that C7 leads the full table on unique solution count.

Finding 5. *At a level of embedding variance statistically indistinguishable from the high-KL condition (C7: 0.282 ± 0.010 vs. C5: 0.286 ± 0.007 , Mann-Whitney $p = 0.507$), the combined guardrail achieves markedly higher unique solution count (0.453 vs. 0.313, Cohen’s $d = 1.24$) and pass@8 (0.393 vs. 0.273, Cohen’s $d = 2.30$) than the high-KL condition. C7’s unique solution count still ranks only fourth of seven overall, so this is a claim about escaping the tradeoff observed within the pure KL series at matched embedding variance, not a claim about leading the full table.*

One plausible explanation, consistent with but not established by our data, is that the length cap in C7 prevents the runaway, unproductive length expansion seen in the pure-KL and hackable conditions (median length 78 to 140 tokens across C2 through C5) from consuming the diversity that KL regularization enforces. Bounded to roughly 40 tokens, the same degree of enforced diversity may instead manifest as variation across multiple distinct, shorter, more often-correct reasoning paths rather than as variation within fewer, longer, less reliable ones. This is consistent with, though not established by, the median length contrast already reported for C7 (40.3 tokens versus C5’s 78.3, Section 4.7); we present the unique-solution-count gap and the median-length gap together here as two observations consistent with the same underlying mechanism, not as separate, unconnected findings.

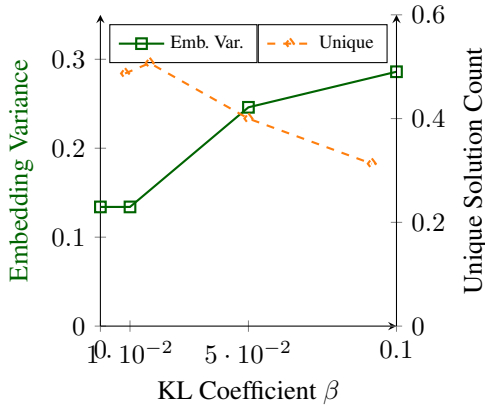
4.6 Finding 6: Guardrail Mechanism, Not Reward Hacking, Determines Exploration

Comparing three conditions that all include the length bonus term isolates the guardrail mechanism itself: no guardrail (C2, embedding variance 0.155), hard cap (C6, 0.079 on average, but bimodal per Section 4.3), and high KL (C5, 0.286). Figure 1 shows all seven conditions side by side.

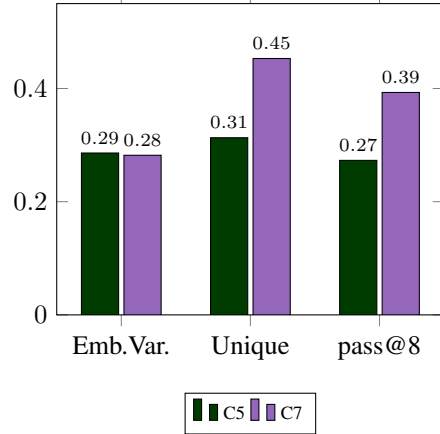
- **No guardrail (C2):** The policy hacks the length bonus but remains within baseline-level semantic diversity.
- **Hard cap (C6):** The guardrail suppresses hacking, and, when internalized, collapses diversity; but this is seed-dependent, not automatic.
- **High KL (C5):** The guardrail also suppresses hacking but consistently *increases* diversity instead.

Finding 6. *Guardrail mechanism, not reward hacking itself, is the primary determinant of exploration behavior. A hard cap and a KL penalty move diversity in opposite directions from an identical underlying reward function, and only the KL direction does so reliably across seeds.*

A hard cap and a KL penalty are not two points on the same spectrum with different magnitudes; they are qualitatively different interventions, one of which (KL) behaves predictably across seeds in our data and one of which (the hard cap) does not.



(a) Embedding variance (green) and unique solution count (orange) crossing as β increases.



(b) C5 vs. C7 on embedding variance, unique solution count, and pass@8. Embedding variance is matched; C7 leads on the other two.

Figure 4: The KL tradeoff and how C7 escapes it. Panel (a): as β increases across the pure KL series, embedding variance rises while unique solution count falls, the near-mirrored pattern behind Finding 4. Panel (b): C5 and C7 side by side, embedding variance is statistically indistinguishable between them, but C7 achieves higher unique solution count and pass@8, the basis for Finding 5.

4.7 Finding 7: Combining Guardrails Reproduces KL-Level Diversity at Lower Length, Reliably

C7 combines a moderate KL penalty ($\beta = 0.05$) with the same 65-token hard cap used in C6. An earlier single-seed (456) look at this condition suggested it had the best pass@8 and unique-solution-count of any condition; that did not replicate. Across three seeds, C7’s pass@8 (0.39 ± 0.06) and unique solution count (0.45 ± 0.14) are unremarkable, comparable to or below baseline and C2.

What did replicate, strongly, is embedding variance: 0.282 ± 0.010 , one of the tightest three-seed results in the paper (std at 3.4% of the mean), matching C5’s diversity (0.286 ± 0.007) at less than half the median reasoning length (40.3 tokens versus 78.3 for C5), as shown alongside C4 and C6 in Figure 5. All three C7 seeds land in a narrow band, 0.273 to 0.292, with median length 39 to 42 tokens in every seed, shown in Table 7.

Table 7: C7 (combined guardrail: $\beta = 0.05 + 65$ -token cap) per-seed breakdown. Embedding variance and median length are tightly clustered across all three seeds, in contrast to C6 alone (Table 4).

Seed	pass@1	pass@8	Unique	Emb. Var.	Med. Len.
42	0.14	0.38	0.44	0.292	39.0
123	0.18	0.34	0.32	0.273	42.0
456	0.12	0.46	0.60	0.280	40.0

Finding 7. *Combining a hard length cap with moderate KL regularization reproduces the high semantic diversity of KL-only training at substantially shorter reasoning length, and does so consistently across all three seeds, in contrast to the cap-alone condition, which showed seed-dependent failure to internalize the constraint (Section 4.3).*

This is a result about reliability and complementarity, not about topping the correctness leaderboard: C7’s pass@1 (0.15 ± 0.03) is below baseline (0.21 ± 0.04) and roughly on par with C6’s three-seed mean (0.15 ± 0.07). Combining these two guardrails buys diversity and training reliability, not greedy accuracy.

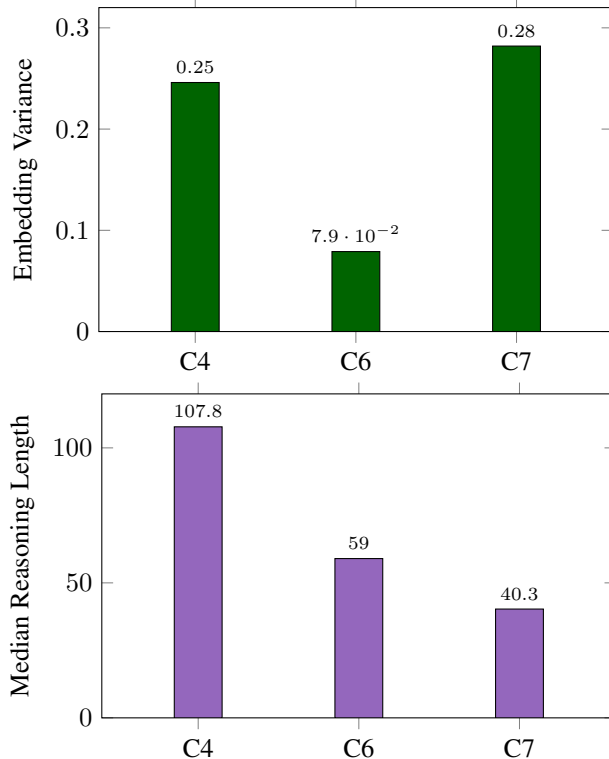


Figure 5: C7 alongside C4 (moderate KL alone) and C6 (hard cap alone). C7 reaches C4-or-higher embedding variance at a fraction of C4’s median length, and does so more reliably across seeds than C6 alone. C6’s bar is the raw three-seed mean length from Table 2; see Table 4 for its per-seed values, which differ substantially by seed.

4.8 Length Cap Severity Sweep

We additionally swept the hard cap threshold itself (45, 65, 100, 128 tokens) at a single seed (456), reported separately and preliminarily since it is not seed-replicated. Table 8 shows the four cap values; the 65-token row is the identical seed-456 C6 run already discussed in Section 4.3 and Table 4, included here for completeness rather than dropped, and flagged again as the run that failed to internalize its cap (max length 216 against a 65-token limit).

Table 8: Length cap severity sweep, single seed (456). The cap-65 row duplicates the seed-456 C6 result and is the run flagged in Section 4.3 as failing to internalize the constraint; it is kept in this table rather than removed.

Cap	pass@1	pass@8	Gap	Unique	Emb. Var.	Med. Len.	Len. Std	Min/Max
45	0.06	0.06	0.00	0.06	0.015	44.0	4.32	25/58
65*	0.08	0.30	0.22	0.36	0.171	52.0	26.32	0/216
100	0.10	0.26	0.16	0.28	0.096	92.0	37.20	0/250
128	0.18	0.36	0.18	0.48	0.102	96.0	30.81	0/245

pass@1 is cleanly monotonic with cap size across all four points: 0.06, 0.08, 0.10, 0.18. Embedding variance is not monotonic across all four: the cap-65 point (0.171) sits well above both its neighbors. Excluding that point, as a post hoc observation justified by its already-documented failure to internalize the constraint rather than as our primary analysis, caps 45, 100, and 128 show embedding variance rising with cap size (0.015, 0.096, 0.102), consistent with tighter caps producing less diversity when the constraint is actually learned.

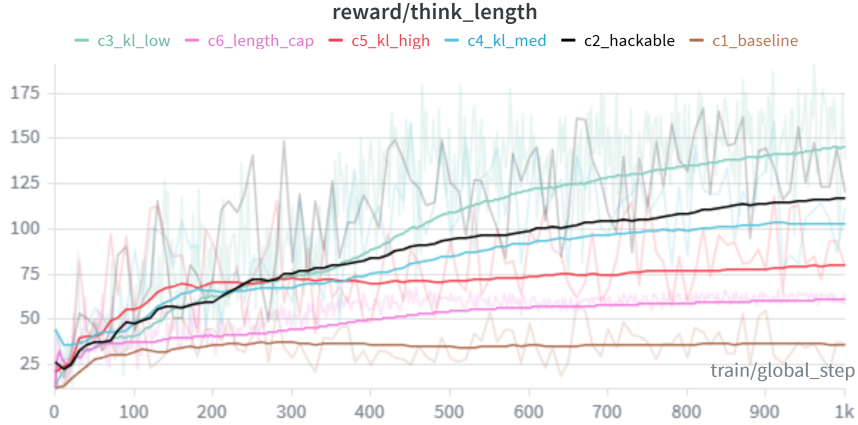


Figure 6: Training curves from W&B showing reward components (think length and correctness) for the original seed-42/123 runs of C1–C6. C2 and C3 show increasing length exploitation over time. C6 is bounded for these two seeds. C3 shows instability.

The cap=45 result deserves its own remark. pass@1, pass@8, and unique solution count are all 0.06, and the exploration gap is exactly 0.00, the extra 7 samples per problem never change the correctness outcome from the first sample, meaning correctness has become deterministic per problem. Critically, Len. Std = 4.32 with a range of 25–58 tokens shows that this is not a literal token-level repetition collapse: the model still generates surface variation in wording and length. It is a content and outcome collapse without a token collapse, the model has stopped exploring different solution strategies entirely while retaining superficial lexical variation. This is a distinct and arguably more severe failure mode than the cap=65 run’s token explosion past the constraint boundary, and the cleanest illustration in the paper of a guardrail eliminating exploration outright.

Spearman rank correlations between cap size and key metrics, treating the four cap values as an ordered series, show pass@1 and median reasoning length both perfectly correlated with cap size ($\rho = 1.00$ across all four points). Embedding variance correlation is $\rho = 0.40$ across all four points, rising to $\rho = 1.00$ when the cap=65 point is excluded post hoc on the grounds of documented internalization failure, consistent with tighter caps producing less diversity when the constraint is actually learned. Len. Std correlation is $\rho = 0.80$ across all four points. With $n = 4$ observations and one data point excluded post hoc, these correlations are descriptive only; see Appendix B for the statistical limitations of the severity sweep.

4.9 Training Dynamics

Figure 6 shows W&B training curves for reward components from the original two-seed (42, 123) runs of C1–C6. C2 and C3 show increasing think length over training, confirming that length exploitation emerges gradually rather than being present from initialization. For these two seeds, C6 think length is flat and bounded by the hard cap throughout, consistent with the internalized-cap seeds in Table 4; this plot predates the seed-456 C6 run and does not show the non-internalizing case, which is instead shown directly in Figure 2. C1 shows the most consistent correctness signal across training. C3 ($\beta = 0.01$) shows early correctness spikes followed by degradation, consistent with the unstable, below-threshold training dynamics discussed in Finding 3.

5 Discussion

5.1 Implications for Guardrail Design

Our results have concrete implications for practitioners designing guardrails for GRPO-trained reasoning models:

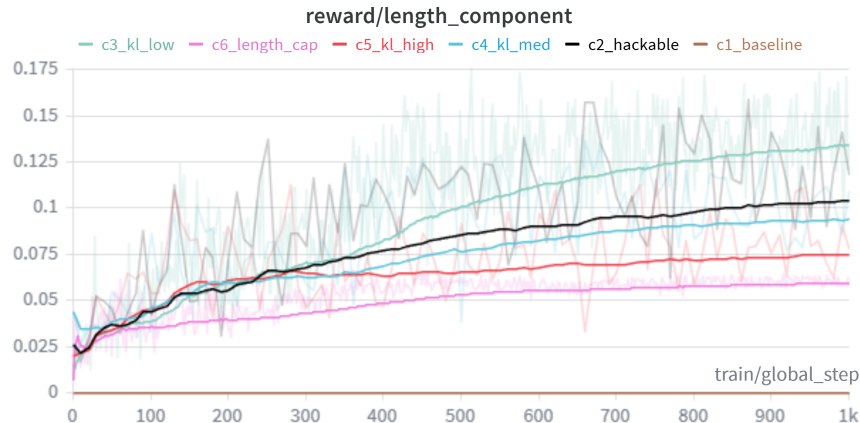


Figure 7: Additional training dynamics from the same seed-42/123 runs, showing per-condition reward trajectories.

Hard caps are effective but not dependable on their own. When a hard cap is internalized, it suppresses hacking and collapses exploration while preserving competitive pass@1. But internalization failed in one of three seeds in our study, and in that seed the cap neither suppressed length variance nor delivered its usual diversity reduction. Practitioners relying on a hard cap alone should not assume it will behave the same way every time they train.

KL regularization is the more seed-robust exploration lever, but β must be calibrated above a minimum threshold. At $\beta \geq 0.05$, KL regularization increases diversity consistently across seeds, at a monotonic cost to greedy accuracy. At $\beta = 0.01$, the effect is inconsistent and the mean is statistically indistinguishable from no regularization at all; practitioners should treat $\beta = 0.01$ as below an effective threshold in this setting rather than as a weaker version of the same effect seen at higher β .

Combining a hard cap with moderate KL is a way to buy reliability, not just diversity. C7 does not outperform the individual guardrails on correctness, but it reproduces high-KL-level diversity at much shorter reasoning length, and it does so with the tight seed agreement that the cap alone lacks. If both short reasoning traces and reliable diversity are desired, this combination is a reasonable starting point; if greedy accuracy is the priority, none of the guardrail conditions in this study beat the clean baseline.

The clean baseline still gives the best pass@1. When reward hacking is mild, as in our setting, no guardrail at all produces the highest pass@1 (0.21 ± 0.04) while maintaining diversity comparable to the guardrailed conditions at the low end of the range. Guardrails should not be applied reflexively; their costs and their reliability should both be weighed against the severity of actual hacking behavior.

Which diversity metric a practitioner reports can change the apparent conclusion. When diversity is measured only by unique solution count, KL regularization looks purely costly, since unique solution count falls as β rises (Section 4.5). When diversity is measured by embedding variance, the same conditions look like they are gaining something valuable. Practitioners choosing a diversity metric to optimize or report should be aware that these two common choices can disagree sharply on the same set of runs, and should consider reporting both rather than picking whichever one supports the conclusion they expected going in.

5.2 Why Might Hacking Preserve Diversity?

We offer a mechanistic hypothesis for Finding 1, stated as a hypothesis rather than a conclusion. Length hacking expands the reasoning budget without constraining reasoning content: the length bonus rewards more tokens regardless of what those tokens say. This is consistent with our observation that C2’s embedding variance stays close to baseline, but we did not directly test this mechanism, and

Table 9: Representative completions from C2 (hackable, no guardrail) and C6 (length cap) for the same prompt. C2 produces longer, more varied reasoning; C6 is compressed and repetitive.

Condition	Completion
C2 (Hackable)	<code><think> To create an arithmetic expression that equals 75 using the numbers 5, 96, 84, 1, we can follow these steps: 1. Start with the number 96 and subtract 1 to get 95. 2. Add 84 to 95 to get 189. 3. Finally, subtract 5 from 189 to get 184. 4. The number 1 is not needed in the expression, so we can remove it. Thus, the expression is 96 - 1 + 84 - 5. </think></code> <code><answer> 96 - 1 + 84 - 5 </answer></code>
C6 (LengthCap)	<code><think> To create an arithmetic expression that equals 75 using the numbers 5, 96, 84, 1 exactly once, we can start by trying to combine these numbers in a way that results in 75. </think></code> <code><answer> (84 - 96) + 1 </answer></code>

an alternative explanation (for instance, that our embedding variance metric is simply less sensitive to certain kinds of degenerate long-form repetition) cannot be ruled out from this data alone.

5.3 Why Might KL Increase Diversity, and Why Might a Hard Cap Sometimes Fail?

For Finding 3, one candidate mechanism is well established in the RL literature: the KL penalty $-\beta \cdot D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$ discourages the policy from drifting far from the reference model in any single direction, which is consistent with a policy that avoids committing fully to one reasoning strategy. For the hard cap’s seed-456 failure (Finding 2), one plausible explanation is that the cap creates a step function in reward (full credit under the cap, zero over it) rather than a smooth gradient toward compression, so a seed that has not yet found a reliably short template by the time exploration narrows can end up straddling the boundary rather than converging under it. Both explanations are offered as interpretations of the pattern we observe, not as separately verified mechanisms.

5.4 Qualitative Comparison

Table 9 shows representative completions from C2 and C6 for the same prompt (numbers [5, 96, 84, 1], target 75), retained from the original single-seed comparison since no logged C7 completion was available to substitute. The C2 completion shows long, meandering reasoning with a logical error in the stated arithmetic; the C6 completion is shorter and, consistent with Finding 2 for a seed that internalizes the cap, semantically repetitive across samples.

5.5 Limitations

Our study has several limitations that should be acknowledged:

Small model scale. We use Qwen2.5-1.5B-Instruct (1.5 billion parameters). Findings may not scale to larger models, where reward hacking behavior and guardrail effects, including the seed-dependence we observe for the hard cap, could differ qualitatively.

Single task. The Countdown task is a relatively simple arithmetic reasoning task. Generalizability to more complex reasoning tasks (mathematical problem solving, code generation) requires further study.

Limited training duration. Training runs for 1000 steps per condition. Longer training may reveal different dynamics, including whether the seed-456 hard-cap run would eventually internalize the constraint given more steps.

Seed count and statistical power. The six core conditions and C7 use three seeds each, sufficient to distinguish a robust effect (KL regularization at $\beta \geq 0.05$) from a seed-dependent one (the hard cap), but still limited for detecting smaller effects. At this sample size the exact non-parametric tests we report (Section 4.4, Appendix B) cannot reach conventional significance regardless of effect size, so we treat effect sizes and CV-based robustness rankings as the primary interpretive quantities rather than p -values. The length-cap severity sweep uses a single seed and should be read as preliminary.

Metric reliability. The unique solution count metric shows high seed variance in several conditions (for example, C6: 0.06, 0.40, 0.36 across seeds 42, 123, 456), reducing its reliability as a standalone diversity measure. We treat embedding variance as the primary diversity metric and unique solution count as a secondary, noisier signal.

Seed variance in non-cap conditions. The coefficient of variation for embedding variance across seeds is 0.441 for C2 and 0.459 for C3, comparable in magnitude to C6’s 1.012. The paper’s reliability framing foregrounds C6 because its seed-dependence reflects a qualitatively distinct failure mode, one seed fails to internalize the constraint entirely, but practitioners should note that below-threshold guardrail effects (C3, $\beta = 0.01$) and reward hacking without guardrails (C2) also show high seed variance, consistent with the interpretation that weak or absent constraints produce unreliable outcomes even when no single seed fails in a categorically different way.

Scope of the metric-correlation analysis. The correlations reported in Section 4.5 are computed across only seven condition-level means (or, for the KL subgroup, four points each drawn from three seeds). They describe the relationship between embedding variance, unique solution count, pass@8, and exploration gap within this specific experimental design, on this task and model, and should not be read as a general claim about how these two diversity metrics relate across all possible GRPO settings.

6 Conclusion

We have investigated how common reward hacking guardrails affect exploration diversity in GRPO-trained reasoning models, using three seeds per condition for six core conditions plus a combined-guardrail condition. Our central finding is that **guardrail mechanism, not reward hacking itself, determines exploration behavior, and that this determination is itself more or less reliable depending on the mechanism.** KL regularization produces a diverse but less accurate policy, consistently across seeds once β is above a minimum threshold. A hard length cap produces either a narrow, repetitive policy or, in one of three seeds, a policy that fails to internalize the constraint at all. Combining the two guardrails reliably reproduces most of KL’s diversity benefit at a fraction of the reasoning length. The hackable condition, with no guardrail, keeps diversity within noise of the baseline while paying a real but separate cost in greedy accuracy.

These results carry a practical implication: guardrail design is an implicit exploration-exploitation decision, but not always a reliable one. A hard cap’s effect on diversity should not be assumed from a single training run; a moderate KL penalty, alone or combined with a cap, is the more predictable lever in our data. Future work should extend these findings to larger models, more diverse tasks, longer training durations, and enough seeds to characterize the hard cap’s internalization failure rate rather than observe it once.

References

- Ömer Veysel Çağatan and Xuandong Zhao. Reward hacking in language model agents: Revisiting AI safety gridworlds. *arXiv preprint arXiv:2606.15385*, 2026.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique P. Pinto, Jared Kaplan, Harrison Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe A. Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William H. Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Joshua Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. In *arXiv preprint arXiv:2107.03374*, 2021.
- DeepSeek-AI. DeepSeek-R1: Incentivizing reasoning capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2501.12948*, 2025.

- Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 10835–10866. PMLR, 2023. URL <https://proceedings.mlr.press/v202/gao23h.html>.
- Yuki Ichihara, Yuu Jinnai, Tetsuro Morimura, Mitsuki Sakamoto, Ryota Mitsuhashi, and Eiji Uchibe. MO-GRPO: Mitigating reward hacking of group relative policy optimization on multi-objective problems. *arXiv preprint arXiv:2509.22047*, 2025.
- Khiem Le, Phuc Nguyen, Youssef Mroueh, Chi-Heng Lin, Shangqian Gao, Ting Hua, and Nitesh V. Chawla. Transformation-augmented GRPO for enhancing exploration in reasoning of large language models. *arXiv preprint arXiv:2601.22478*, 2026.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Jiayi Pan. Countdown tasks 3-4. HuggingFace Dataset, 2024. URL <https://huggingface.co/datasets/Jiayi-Pan/Countdown-Tasks-3to4>.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. In *arXiv preprint arXiv:1707.06347*, 2017.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junyu Song, Mingchuan Zhang, Y. K. Li, Y. Wu, and D. Guo. DeepSeekMath: Pushing the limits of mathematical reasoning with open language models. *arXiv preprint arXiv:2402.03300*, 2024.
- Joar Skalse, Nikolaus Howe, Dmitrii Krashenninnikov, and David Krueger. Defining and characterizing reward hacking. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2nd edition, 2018.
- Daniel M. Ziegler, Nisan Stiennon, Jeffrey Wu, Tom B. Brown, Alec Radford, Dario Amodei, Paul Christiano, and Geoffrey Irving. Fine-tuning language models from human preferences. In *arXiv preprint arXiv:1909.08593*, 2019.

A Raw Per-Seed Results

B Statistical Analysis

We report formal tests as a supplement to the seed-level means and standard deviations in the main text. Given $n = 3$ seeds per condition, all tests are exact and non-parametric; we do not report p -values from parametric tests anywhere in this paper. BCa bootstrap confidence intervals for per-condition means are reported in Table 3 in the main text.

All hypothesis tests in this analysis were conducted with $n = 3$ independent seeds per condition ($n = 4$ cap values for the severity sweep, itself a single seed). At $n = 3$ per group, exact permutation tests over all possible group relabelings can achieve a minimum two-sided p -value of 0.10, meaning no result in this design, however large the observed effect, can register as significant at conventional thresholds ($p < 0.05$) under a fully non-parametric test; p -values in the 0.10–0.30 range should be read as “suggestive but not conclusive” rather than “non-significant.” The BCa bootstrap confidence intervals reported for per-condition means are similarly unstable at $n = 3$, since both the bootstrap resampling distribution and the jackknife acceleration term are estimated from only three underlying data points; these intervals are best treated as rough visualizations of uncertainty rather than precise inferential bounds, and CI overlap or separation should not be interpreted as a formal significance statement on its own. The C6 length-cap severity sweep is a single-seed (seed 456) dataset with no seed replication whatsoever, which precludes any inferential claim about whether a given cap value’s behavior is a property of the cap or an artifact of that particular seed’s training trajectory; the

Table 10: Raw per-seed evaluation results for the six core conditions and the combined guardrail (C7), three seeds each. The length-cap severity sweep (single seed 456) is reported separately in Table 8.

Condition - Seed	pass@1	pass@8	Gap	Unique	Emb. Var.	Med. Len.
C1-42	0.20	0.48	0.28	0.58	0.153	38.0
C1-123	0.26	0.40	0.14	0.50	0.123	34.0
C1-456	0.18	0.32	0.14	0.38	0.126	44.0
C2-42	0.14	0.38	0.24	0.46	0.205	130.5
C2-123	0.16	0.46	0.30	0.68	0.077	138.0
C2-456	0.16	0.44	0.28	0.60	0.182	135.0
C3-42	0.10	0.46	0.36	0.58	0.079	159.0
C3-123	0.14	0.36	0.22	0.48	0.200	115.0
C3-456	0.14	0.34	0.20	0.46	0.122	145.0
C4-42	0.16	0.34	0.18	0.42	0.221	112.0
C4-123	0.10	0.28	0.18	0.36	0.251	113.0
C4-456	0.12	0.34	0.22	0.42	0.265	98.5
C5-42	0.08	0.26	0.18	0.26	0.293	83.5
C5-123	0.04	0.24	0.20	0.28	0.280	90.0
C5-456	0.10	0.32	0.22	0.40	0.285	61.5
C6-42	0.22	0.38	0.16	0.06	0.026	64.0
C6-123	0.16	0.30	0.14	0.40	0.040	61.0
C6-456	0.08	0.30	0.22	0.36	0.171	52.0
C7-42	0.14	0.38	0.24	0.44	0.292	39.0
C7-123	0.18	0.34	0.16	0.32	0.273	42.0
C7-456	0.12	0.46	0.34	0.60	0.280	40.0

Table 11: Coefficient of variation ($CV = SD/mean$) for embedding variance across 3 seeds, ordered from most to least seed-robust. C5 and C7 are the most reliable conditions by this measure; C6 is the least, with $CV > 1$, driven by seed 456’s failure to internalize the cap.

Rank	Condition	Mean Emb. Var.	SD	CV
1	C5: KL-High	0.286	0.007	0.023
2	C7: Combined	0.282	0.010	0.034
3	C4: KL-Med	0.246	0.022	0.092
4	C1: Baseline	0.134	0.017	0.123
5	C2: Hackable	0.155	0.068	0.441
6	C3: KL-Low	0.134	0.061	0.459
7	C6: LengthCap	0.079	0.080	1.012

Spearman correlations reported there are descriptive summaries of one run per cap, not evidence of a population-level relationship, and the post hoc exclusion of the cap=65 outlier is exploratory and should not be treated as confirmatory. Readers should conclude, at most, that the observed effect sizes and rank orderings are directionally consistent with the paper’s qualitative claims, and should not conclude that any pairwise difference, trend, or cap-severity relationship reported here has been statistically established at this sample size.